Big Data Technologies

Infrastructure Health Analysis for Digital Hospitals using the Hadoop Ecosystem

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Abstract

Modern hospitals generate millions of machine-generated logs every hour from Wi-Fi routers, servers, and digital medical devices such as MRI and X-ray systems. Due to their volume and velocity, these logs cannot be efficiently analyzed using traditional database systems. This project implements a Hadoop-based Big Data framework to store and analyze these digital footprints. Apache HDFS is used for scalable and fault-tolerant storage, while Apache Hive enables large-scale analytical querying. The system identifies security threats such as brute-force attacks and detects abnormal machine error patterns, helping improve both network security and operational reliability in hospital environments.

Introduction

The Problem:
Hospitals are high-value targets for cyberattacks such as ransomware. The massive volume of system-generated logs makes manual monitoring and analysis impractical.

The Data:
The project focuses on analyzing system logs, which are automatically generated text files produced by servers, network devices, and medical equipment.

The Solution:
A distributed log analysis framework using Hadoop. Instead of processing logs on a single machine, the system uses a cluster-based approach to detect anomalous patterns across large datasets.

Objective Overview:
To build a scalable pipeline that ingests, cleans, stores, and queries system logs to identify security threats and operational issues.

Objectives

- Design a Scalable Data Warehouse: To create a storage system using HDFS capable of storing large volumes of hospital network logs reliably.
- Automate Data Cleaning: To develop MapReduce programs that preprocess raw machine logs and extract relevant security and operational fields.
- Identify Security Threats: To use Apache Hive for writing analytical queries that detect brute-force attacks and abnormal access patterns through historical log analysis.
- Establish a Reliable Data Pipeline: To demonstrate end-to-end data flow from hospital databases into a Big Data environment using Apache Sqoop.
- Improve Operational Reliability: To identify early warning signs of medical equipment malfunction by analyzing recurring error logs.

Literature Survey

S.No	Research Paper (Title & Link)	Author(s)	Drawbacks / Gaps	Future Directions
1	**Big Data for Secure Healthcare System: A Conceptual Design** https://link.springer.com/article/10.1007/s40747-017-0040-1	Research Team (Secure Healthcare Study)	Conceptual model only; no Hadoop ecosystem implementation.	Implement full Hadoop pipeline (Sqoop → HDFS → MapReduce → Hive).
2	**Log Analysis with Hadoop MapReduce** https://dergipark.org.tr/en/pub/ject/article/980060	Gligor Risteski et al.	Uses traditional MapReduce; high latency for alert generation.	Replace batch processing with Hive/Spark SQL for faster querying.
3	**Enhanced Secured MapReduce Layer for Big Data Privacy and Security** https://doi.org/10.1186/s40537-019-0193-4	P. Jain, M. Gyanchandani, N. Khare	Focused on framework-level security; lacks practical log-analysis case study.	Integrate security layer with healthcare log analytics pipeline.
4	**Digital Forensics Readiness in Big Data Networks: A Framework for Linux–Hadoop Environments** https://doi.org/10.3390/asi7050090	Cephas Mpungu et al.	Emphasizes forensic response after attack, not proactive threat detection.	Develop proactive anomaly detection queries using Hive.
5	**A Novel Hadoop Security Model for Addressing Malicious Collusive Workers** https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8285196/	(PMC Research Group Authors)	Focused only on cluster-node security, not log-based intrusion detection.	Combine node-level protection with log anomaly detection.

6	**Evading Cyber-Attacks on Hadoop Ecosystem: ML-Based Security-Centric Approach** https://doi.org/10.3390/info15090558	Neeraj A. Sharma et al.	Heavy ML dependency; requires high computational resources.	Implement lightweight rule-based Hive queries for smaller hospitals.
7	**Aadhaar Data Analysis Comparison in MapReduce, Hive and Spark** https://doi.org/10.2991/ahis.k.210913.036	R. Roopa et al.	Not security-focused; dataset unrelated to cybersecurity logs.	Apply comparative analysis specifically to hospital system logs.
8	**Big Data Analytics in Healthcare: Data Sources, Tools, Challenges and Opportunities** https://doi.org/10.1186/s43067-024-00190-w	Multiple Authors (Springer Open Survey)	Broad survey; lacks technical implementation details.	Develop practical Hadoop-based implementation for hospital infrastructure.
9	**Analyzing Web Access Logs using Spark with Hadoop** https://ijcaonline.org/archives/volume180/number1/28768-2017915904/	Vandita Jain et al.	Focused on web logs; not healthcare or ransomware detection.	Extend approach to medical device and network router logs.
`	**Big Data for Secure Healthcare System: A Conceptual Design** https://link.springer.com/article/10.1007/s40747-017-0040-1	Research Team (Secure Healthcare Study)	Conceptual model only; no Hadoop ecosystem implementation.	Implement full Hadoop pipeline (Sqoop → HDFS → MapReduce → Hive).

Conclusion

- **Successful Scalability:** The project proves that Hadoop can process millions of machine logs that would cannot be processed efficiently by traditional systems.
- **Enhanced Security:** We successfully identified simulated "Brute Force" attacks by detecting high-frequency login failures in the logs.
- **Operational Efficiency:** By monitoring "Error Logs," the system can alert IT staff about medical equipment failure (like an X-ray machine overheating) before it stops working.

THANK YOU